METHODS NOTE · PHONES WITHOUT ACCOUNTS

How the figures were produced, and how far they can be trusted

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Companion to the two-page insight brief · All queries in queries.sql

Question

Among adults in the areas Findex surveyed in Ethiopia who already own a mobile phone, how many have no account, who are they, and why do they say they have none?

Data

- World Bank Global Findex 2025, individual microdata, CSV release (144,090 respondents, 141 economies). Ethiopia: 1,001 adults aged 15+, face-to-face interviews, 15 Oct–29 Nov 2024.
- **Coverage:** the World Bank excluded Amhara (state of emergency), Harari, Somali and Tigray, about 30% of the population. All figures describe the covered areas only, and no national headcount is estimated.
- Loaded into PostgreSQL, filtered to Ethiopia, then cast to a typed working table (the import read most columns as text).

Definitions

- **Has a phone:** `con1 = 1` (own phone for personal calls). One "don't know" answer is excluded from phone figures.
- **Has an account:** `account = 1`, the World Bank's combined measure (bank or similar institution, or mobile money).
- **Basic phone:** `con9 = 2`; asked only of phone owners.
- **Reasons:** `fin11a–f`, asked of everyone without a bank-type account. Several can be "yes", so they don't add to 100.

Weights

Every figure uses the survey weight `wgt`. For Ethiopia the weights sum to 1,001, so a share is simply the weighted count over the weighted total.

Checks before analysis

Check	Result
Global row count	144,090
Ethiopia rows	1,001
Missing weights (Ethiopia)	0
Weight range	0.21 – 2.94
Account ownership, unweighted	58.7%
Account ownership, weighted	**48.8%**
Published Findex figure	48.8%
Four groups add to	100.0%

The weighted figure matching the published one confirms the load, the filter and the weighting. Skipping the weights would have overstated account ownership by 10 points, because the raw sample over-represents adults who have accounts.

Uncertainty

Intervals are approximate 95% margins: $1.96 \times \sqrt{p(1-p)/n_{eff}}$, where $n_{eff} = (\Sigma w)^2 / \Sigma w^2$ (591 for all adults, 405 for phone owners). Check: $n_{eff} = 591$ implies a design effect of 1,001 / 591 = 1.69, exactly the World Bank's published figure for Ethiopia, whose overall margin is ±4 points. Like the World Bank's, this reflects weighting only, not clustering, so subgroup intervals are approximate and are not used to call small differences significant. Groups under 60 respondents are flagged.

What I chose to show, and why

- **Waffle of 100 adults** for the headline: a part-of-whole read in people, not percentages, with one highlight colour for the group the brief is about.
- **Dot plot with intervals** for who is affected: it puts the finding (clear gaps by schooling, place and age; no clear pattern by income) and its uncertainty in the same picture, so the claim can't outrun the data.
- **Sorted bars** for reasons, with the leading reason highlighted.
- Left out: splits such as employment status and gender, where differences were small or the sample too thin to support a claim.

Limits

- Cross-tabs show patterns, not causes. Education, rurality, age and income overlap; separating them needs a regression, which is a sensible next step.
- Reasons are self-reported and asked of those without a bank-type account. 15–17 year olds may face age rules for opening accounts.
- One survey round; no trend over time for the connectivity questions, which are new in 2025.

Tools

PostgreSQL (DataGrip) for loading, checks and every reported figure; Python for the charts and layout. I used an AI assistant (Claude) for drafting queries, chart code and copy. Every figure was recomputed in SQL and the totals checked against the published Findex number before anything went into the brief.